

August 28, 2025

Andrew S. Johnston
Executive Secretary
Maryland Public Service Commission
William Donald Schaefer Tower
6 St. Paul Street, 16th Floor
Baltimore, Maryland 21202

Re: Case No. 9761; Comments of VGIC on Utility VPP Programs

Dear Mr. Johnston,

The Vehicle-Grid Integration Council (“VGIC”)¹ respectfully provides its comments in response to the Baltimore Gas and Electric (“BGE”), Potomac Edison Company (“Potomac Edison”), and Potomac Electric Power Company’s (“PHI”) and Delmarva Power & Light Company (“Delmarva Power”; collectively with PHI the “PHI Utilities”) applications for Distribution System Support Services (“DSSS”) Pilots to the Maryland Public Service Commission (“Commission”) in Case No. 9761.

I. Introduction

VGIC is a 501(c)(6) membership-based advocacy group committed to advancing the role of electric vehicles (“EVs”) and vehicle-grid integration (“VGI”) through policy development, education, outreach, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring the value from EV deployments and flexible EV charging and discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid.

VGIC appreciates the BGE, Potomac Edison’s, and the PHI Utilities proposals to advance virtual power plants (“VPP”) and grid-parallel bidirectional charging (often referred to as vehicle-to-grid or “V2G”) through the DSSS pilot applications. These utilities have been leaders in making the transition to electric vehicles affordable, reliable, and clean.

When the DRIVE Act was enacted in spring 2024, VGIC was hopeful that it would lead to meaningful opportunities for grid-parallel bidirectional charging systems or “V2G” systems, as well as storage-paired unidirectional charging systems, to support the grid and, in return, be recognized and compensated for grid value. On June 11, 2025, the Commission unanimously

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, BMW, BodeEV, BorgWarner, ChargePoint, ChargeScape, Codibly, Critical Loop, dcbel, Eaton Corporation, Emporia Corp., EnergyHub, EV.Energy, Fermata Energy, Gravity, GridBeyond, GridWiz, Innovation Core SEI, Kaluza, Landis + Gyr, LG Innotek, Mercedes-Benz, Orange Charger, Rivian, Stellantis N.V., Sunrun, Tesla, The Mobility House, Toyota Motor North America, Inc., UL Solutions, WeaveGrid, Clean Power Alliance, Holy Cross Energy, Los Angeles Department of Water & Power, MCE Energy, San Diego Community Power, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>)

adopted expedited V2G interconnection processes for all utilities in the state, addressing a significant barrier to the widespread use of V2G solutions to support the grid. VGIC appreciates the focus of the Commission and Commission staff in efficiently and fairly implementing these new interconnection rules. VGIC commends the utilities for continuing progress with the DSSS pilot applications, filed July 1, 2025. These pilots, if implemented effectively, are critical to realizing Maryland's commitment to clean and the electrification of transportation. The pilots could aid in achieving Maryland's vision that 100% of passenger car and light truck sales be EVs by 2035. The DSSS pilots also hold the potential to further the policy goals of the Maryland Climate Solutions Now Act of 2022, which targets a 60% reduction in greenhouse gas emissions by 2031, along with a goal to reach net-zero by 2045.

VGIC provides the following recommended enhancements to each utilities' filings to ensure the pilots can enable V2G resources and VPPs to improve grid flexibility, lower ratepayer costs, and support broader climate goals. These recommended enhancements are summarized below:

- VGIC recommends the utilities detail an “on-ramp” that leverages the pilot learnings to unlock larger program customer count, MW targets, and budgets without disrupting customer participation.
- VGIC advises against prohibiting injection into the distribution grid (i.e., net export), as Maryland's V2G interconnection rules specifically ensure grid safety.
- Given solar customer propensity for adopting bidirectional charging solutions, VGIC urges the utilities to clearly define how Net Energy Metering (“NEM”) systems will be treated in the DSSS pilots.
- VGIC recommends utilities include a locational peak relief component to the pilots from the outset.

II. VGIC recommends the utilities detail an “on-ramp” that leverages the pilot learnings to unlock larger program customer count, MW targets, and budgets.

VGIC is confident that the authors and supporters of the DRIVE Act had envisioned larger participation of bidirectional charging customers in grid support efforts than what is detailed in the DSSS pilot applications. For example, BGE's application targets only a few dozen bidirectional charging customers over the course of the pilot. This scale is also significantly dwarfed by similar VPP initiatives, for example those offered by utilities in Massachusetts, California, Vermont, and Utah. VGIC believes the utility DSSS pilots should be offered to more customers and authorized with a larger budget to provide critical market development support for bidirectional charging solutions. Critically, these V2G technologies do not have access to any other performance-based compensation mechanisms in Maryland. Moreover, these solutions are extremely limited in the opportunities to participate in grid support programs around the country, which means that the Commission could take decisive action to emerge as the definitive national and even global leader in V2G market development. While the technical capabilities of these solutions have been demonstrated repeatedly, including in Maryland, the commercial ecosystem

is relatively nascent. This means that customers can expect equipment and installation cost declines over the coming years.

With these impending cost advantages and the opportunity to establish pioneering V2G pilots, Maryland can ensure that sufficient V2G capacity is available in the long term to serve the state's evolving energy needs. The availability of low-cost, grid-edge energy storage solutions underpinned by the sheer scale of EV adoption can meaningfully mitigate the need for increased investments in generation, transmission, and distribution infrastructure to support the state's growing demand without compromising affordability for all utility ratepayers. In this way, V2G will grow to become an increasingly critical tool in the energy planning toolkit for an affordable, clean future. VGIC respectfully urges the Commission to leverage the DSSS pilots as an opportunity to attract customers and solution providers to the state, and ensure utilities gain familiarity with a solution set that will be key to unlocking Maryland's state transportation electrification and broader decarbonization goals.

While VGIC reiterates the value of providing large-scale V2G offerings to customers, we are sensitive to the increased pressure on Maryland utility ratepayers and understand the need to right-size the DSSS pilots to protect against outsized customer bill impacts for non-participants.

With this in mind, VGIC urges the Commission to direct the utilities to, at a minimum, detail an *on-ramp* plan to scale the pilots, if successful, to mass-market offerings.

Specifically, utilities should provide clear customer enrollment triggers for when consideration of extended pilots and/or mass-market program offerings would be requested. Without a clear plan that bridges pilot implementation to larger-scale program development, the DSSS pilots could become yet another example of "death by pilots," wherein technologies that have been demonstrated to work, i.e., V2G charging systems, get stuck in utility pilots that more closely resemble science fair projects than earnest efforts to learn about customer enrollment strategies, resource performance, and the ratepayer impact of programs. Moreover, requesting pilot extension or a transition into full-scale programs only after an initial pilot concludes is an effective strategy to disrupt customer participation and lose customer engagement. Without continuity for these early adopters and engaged customers, the ultimate goals of the DRIVE Act, Maryland's 2035 transportation electrification ambitions, and the Climate Solutions Now Act of 2022 would be undermined.

III. VGIC advises against prohibiting injection into the distribution grid (i.e., net export), as Maryland's V2G interconnection rules specifically ensure grid safety.

The list of readily available light-duty bidirectional charging systems continues to grow, with offerings from Ford (F-150 Lightning Electric with Ford Charging Station Pro), GM (nearly every Chevrolet, Cadillac, and GMC electric vehicle with GM Energy PowerShift Charger), Nissan (LEAF and dcbeL Ara), Tesla (Cybertruck and Universal Wall Connector), and Kia (EV9 and Wallbox Quasar 2). These solutions are primarily marketed and deployed for islanded back-up power use cases, as opposed to grid-parallel use cases. However, in 2024, BGE became the first utility in the nation to implement grid-parallel residential bidirectional charging configurations.² Additional product announcements have revealed commitments to bidirectional

² <https://investors.sunrun.com/news-events/press-releases/detail/318/sunrun-launches-nations-first-vehicle-to-home-grid>

charging solutions from Volvo, Volkswagen, Hyundai, Polestar, BMW, Mercedes-Benz, Stellantis, Rivian, Lucid, Emporia, ChargePoint, Enphase, SolarEdge, and Autel, among others. Based on conversations with its 40+ members, including leaders in the residential bidirectional charging space, VGIC can attest to significant manufacturer interest in deploying more bidirectional charging systems in safe, utility-interconnected grid-parallel configurations. As such, we believe there will be ample equipment available to provide safe injection onto the grid that expands the benefits of bidirectional charging equipment beyond just meeting site load.

With this in mind, and given the Commission’s recent unanimous approval of V2G interconnection rules, we were surprised to see the utility DSSS pilot applications focus on the use of bidirectional charging equipment to meet site load rather than export to the grid. Additionally, the size of vehicle batteries and chargers dwarf typical home load, which means limiting performance to the size of residential site load is a significant restriction on the technical grid support potential of V2G. For context, VGIC asserts there are four key barriers to widespread bidirectional charging market development:

(1) Interconnection: i.e., *“Will my utility know what to do with an interconnection application for this solution?”*

(2) Compensation: i.e., *“Will I have the opportunity to lower my bill or maybe even receive payments from my utility?”*

(3) Upfront Cost: i.e., *“Will the system be affordable?”*

(4) Technical standards: i.e., *“Will the system be safe and reliable?”*

COMAR Rulemaking Session 203, held on June 11, 2025, addressed the interconnection barrier by ensuring V2G systems are appropriately incorporated into the utilities’ interconnection processes. To the degree there are any concerns from the utilities regarding grid impacts associated with exports, the interconnection review process is specifically intended to be able to evaluate and mitigate such impacts. In other words, not only are UL-certified V2G systems eligible for interconnection under the new rules, but the rules themselves ensure that the impacts of exporting can and will be evaluated and, if those impacts are found to be adverse, mitigations can be required.

The second barrier, compensation mechanisms, would be addressed through the performance-based compensation detailed in the DSSS pilot applications. The third barrier, upfront cost, is partially addressed by both initiatives. The clarity around V2G interconnection pathways, particularly for V2G AC systems (where the bidirectional power conversion functionality is onboard the vehicle), provides a path toward lower-cost, mass-market bidirectional charging systems beyond today’s “first generation” offerings from the prominent manufacturers. Meanwhile, the DSSS pilot designs can help offset upfront cost reductions. For example, Potomac Edison offers a \$150/year “Connectivity Incentive” that is not based on performance during test or real events, but rather on the connection of the device into the program. The final barrier, technical standards, was partially addressed in the interconnection rules, which provides clear guidance on the required safety and grid interoperability requirements of bidirectional

charging equipment. Additional technical standards development is being completed by industry-led working groups and, although supportive of customer adoption, is not in scope for utilities or the Commission (e.g., physical connectors and/or communications between vehicles and chargers).

With these barriers largely addressed by the recent interconnection update and the proposed DSSS pilots, VGIC notes that restricting the performance of V2G solutions to meeting site load, rather than providing net export, is not based in any apparent market development barriers.

Although the utilities do not provide a detailed explanation as to why this limitation will be placed, the DSSS pilot applications do reference the need for a high degree of visibility at the grid edge to manage exports. While investments to expand the utilities' visibility and management capabilities can be valuable, VGIC disagrees that extensive investments, beyond what is needed to enable the relevant program administrator(s)/aggregator(s) envisioned to implement the pilot, are *required* to allow net export onto the grid as part of the DSSS pilots. We reiterate that Maryland's robust interconnection rules already provides – in fact, requires – an opportunity for every site to be evaluated and for any material impacts to be mitigated during the interconnection process.

The proposed DSSS pilot structures are relatively simple, and VGIC fails to understand why such event-based program designs, intended to relieve stress on the grid during peak times, should express a preference for customers that reduce site load over those that reduce site load *and* export to the grid. However, VGIC remains sensitive to the affordability concerns that new programs can raise for Maryland ratepayers. With this in mind, we note that the customer enrollment and budget caps that the Commission will adopt for the DSSS pilots already provide protection against undue upward rate pressure. Further restricting the pilot to only pursue non-net-exporting customers as a means to limit program spending would be duplicative of any program budget or customer caps.

Lastly, authorizing DSSS pilots to advance VPP and V2G without directing utilities to allow enrollment of net exporting customers sends an extremely mixed regulatory signal to customers and the industry. The DRIVE Act envisions meaningful support of the grid through innovative VPP and V2G offerings. Excluding participation of net exporting customers would undermine the intent of the DRIVE Act and, in turn, present a barrier to the achievement of Maryland's broader decarbonization and affordability goals.

IV. Given solar customer propensity for adopting bidirectional charging solutions, VGIC urges the utilities to clearly define how Net Energy Metering (“NEM”) systems will be treated in the DSSS pilots.

There is a strong correlation between rooftop solar ownership and EV ownership, with some studies showing that 25% of EV owners also own rooftop solar, whereas only 8% of non-EV owners have rooftop solar.³ VGIC expects the overlap between solar customers and those adopting bidirectional charging solutions to be even higher. BGE details its intent to focus on

³ <https://www.nrel.gov/news/detail/press/2024/news-release-green-wheels-bright-skies-nrel-analysis-unveils-the-connection-between-electric-vehicles-and-photovoltaics>

customers with solar, stating “specifically, BGE will focus on two DER classes – stationary energy storage devices and bidirectional-enabled EVs – with an emphasis on customers that have paired those DERs with rooftop, BTM solar...”⁴ However, BGE’s filing, as well as the applications of the other utilities, contain a material omission by not detailing the treatment for these customers. Most likely, new or existing solar customers will take advantage of Maryland’s net energy metering (“NEM”) policies to unlock compensation to pay down the upfront cost of the resource. VGIC offers below two options for ensuring these NEM customers can participate in the DSSS pilots without compromising their NEM payments:

- A. Leverage embedded data collection solutions in the V2G system.** Device-level measurement of V2G equipment can provide the utility with the required data to reward performance under the DSSS pilot scheme. This same data can also be subtracted from any utility meter data before NEM credits are rewarded to the customer to ensure that NEM credits are not being rewarded to the V2G system. Given the relatively limited scale of the initial DSSS pilots, VGIC believes this process can be done manually at first, and the utilities can transition to an automated process as enrollment increases, which is a process that’s been directed in California.
- B. [Not recommended] Remove customers from NEM seasonally or for the duration of the pilot.** VGIC advises against pausing NEM for participating customers, as this will negatively impact customer experience and may severely limit the overall number of customers interested in enrolling in the DSSS pilots.

VGIC urges the utilities to clearly define how new and existing rooftop solar systems taking service under NEM will be treated for the purposes of these VPP pilots. Proceeding without a clear plan for treating these customers would severely limit pilot participation and useful learnings. Additionally, VGIC notes that other approved and implemented VPP pilots and programs around the country, including California’s Demand Side Grid Support (DSGS) Program, and the Aggregator VPP program settlement recently filed in Colorado, do not exclude NEM customers from participation.

V. VGIC recommends utilities include a locational peak relief component to the pilots from the outset.

As adoption of EVs and other grid-edge solutions accelerates – and utilities continue to be faced with new and increasingly complex grid management challenges from other trends – the utilities should be encouraged to pursue locational peak relief component from the outset of the DSSS pilots, rather than deferring to later pilot or program design. That said, this should be done in whatever manner is deemed most feasible for the utilities. Implementation complexity and cost for unlocking location-specific distribution system value can vary, as evident in the following examples:

- A. Programmatic.** In this approach, programmatic options focused on distribution optimization offer an opportunity to provide locational peak relief and, in exchange, customers receive participation incentives. For example, BGE and the PHI utilities are advancing the use of managed charging platforms to reduce the strain on

⁴ BGE DSSS Pilot Application. Page 11.

distribution assets. The DSSS pilots could be configured so that events are called when distribution assets are expected to have limited capacity. In this way, events are targeted strictly to distribution assets that are capacity-constrained.

Under the PHI Utilities' DSSS pilot incentive design, customers would only be compensated when they participate in a distribution-based event, meaning that incentives would only be provided to participants who directly benefit distribution assets.⁵ For BGE's incentive design, participants should be compensated based on their expected contribution to the distribution system.⁶

A program could designate that a bulk system event and distribution event cannot be called on the same day. Participants should receive an additional incentive for participating in distribution-based events

- B. **NY's VDER Tariff – Locational System Relief Value (LSRV).** Rather than a programmatic approach (i.e., with a program administrator/aggregator), this approach leverages an open tariff that customers can enroll in. The VDER Tariff simply does or doesn't apply a locational value based on whether the customer is located in a constrained area as indicated on the utilities' hosting capacity maps. In other words, a customer is assigned to one of two representative circuits: "constrained" or "unconstrained." A location-specific adder is then provided to customers in the constrained areas.

Although Maryland utilities do maintain hosting capacity maps, VGIC notes an example of simple distribution value treatment that does not rely on updated hosting capacity maps: Massachusetts' Clean Peak Energy Standard's *Distribution Circuit Multiplier*.⁷ In this approach, the Massachusetts Department of Energy Resources leverages the utility's annual distribution planning data to rank circuits using a three year average of the peak-to-normal percent rating for each circuit. Customers on the 10% of circuits with the highest peak-to-normal rating are then eligible for the distribution element.

Yet another approach could be to award distribution value only to those customers on circuits with peaks that are coincident with system peaks. This is the methodology included in the recent settlement filed by Xcel Energy and several other stakeholders in a settlement being considered by the Colorado Public Utilities Commission. Parties agreed that the utility's avoided distribution capacity value was \$71.93/kW-yr. They agree that the distribution component should be made available to customers on circuits with a historical hourly peak between 4-10pm in June-September.

- C. **PG&E's V2X Pilot** assigns each customer to a "representative circuit" most similar to that customer's circuit load shape. This reduces complexity by obviating the need

⁵ BGE DSSS Pilot Application. Page 19.

⁶ PHI Utilities DSSS Pilot Application. Page 8.

⁷ <https://www.mass.gov/doc/cps-distribution-circuit-multiplier-guideline/download>

to analyze – in the case of PG&E – thousands of distribution circuits. Instead, customers are assigned to one of just a few dozen representative circuits. A location-specific signal is then generated for these representative circuits.

- D. **PG&E’s Flex Connect.** This pilot uses a utility DERMS platform to implement dynamic import limits based on near-real-time, locationally-specific circuit conditions. Note that the Flex Connect program is not linked to a compensation mechanism, but rather is focused on the flexible service connection use case for new large load connections.

VGIC recommends the Commission direct utilities to implement a location-specific distribution element that best suits the current capabilities of each utility. Otherwise, the pilot learnings will be limited to overall system load shift which, while valuable, misses the opportunity to better understand the implications of a more granular focus on utility operations, customer engagement, and, ultimately, utility ratepayers.

VI. Conclusion

The utility DSSS pilot applications all represent significant advancements in leveraging the electrification of transportation in a way that creates benefits for all Maryland ratepayers and residents. VGIC recommends the Commission approve these pilots with the modifications we propose above, to ensure Maryland continues to lead on grid reliability, energy affordability, and V2G market development.

VGIC appreciates the opportunity to provide comments and looks forward to further collaboration with stakeholders on these important initiatives.

Respectfully submitted,
/s/ Zach Woogen
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